Indian Institute of Information Technology Kottayam

StereoDetect
Explainable Stereotype Detection

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Introduction

Five labels capture different forms of group-related language

Why this task matters

Group references and generalizations require context-sensitive classification.

Project focus

Compare model predictions and inspect the words influencing their decisions.

0	1	2	3	4
Anti-stereotype	Stereotype	Neutral without target term	Neutral with target term	Bias

Dataset definitions: StereoDetect [1]. Labels match the supplied CSV annotations.

Literature Review

Dataset foundations and explainable stereotype detection

StereoDetect · Shejole & Bhattacharyya

Definition-aligned benchmark for stereotypes, anti-stereotypes, neutrality, and bias.

Provides the five-class labels used in this project. [1]

HEARTS · King et al.

Combines ALBERT-based stereotype detection with LIME/SHAP explanations.

Motivates diagnostic word attribution for our five-class setting. [2]

The HEARTS binary task and StereoDetect five-class task use different evaluation settings.

Literature Review

Model families used in the comparison

Model family	Core idea	Role in this project
MiniLM [5], [6]	Distilled transformer embeddings	Compact baseline
Qwen3-Embedding [7]	Decoder-derived text embeddings	Embedding baseline
ALBERT [3]	Factorized embeddings; shared layers	Sentence classifier
ModernBERT-large [4]	Local/global encoder attention	Sentence classifier
Qwen3.5-2B + LoRA [8], [9]	Hybrid decoder; task adaptation	Label or JSON output

Architectures: original papers and official model cards [3]–[9].

Motivation

Test both classification quality and the usefulness of the output

1 · Adaptation

How much does task-specific fine-tuning improve the existing Qwen checkpoint?

2 · Output scope

Can one sentence produce the category, target group, and label together?

3 · Explanation

Which words influence an encoder decision, including a wrong prediction?

Project questions evaluated on the supplied StereoDetect splits.

Problem Statement

Two Qwen task contracts, evaluated on the same held-out test split

Task A · Label prediction

Input: Category + Target + Sentence

Output: one integer label, 0–4

Decision: highest-scoring label token

Task B · All-field prediction

Input: Sentence only

Output: category + target + label JSON

Decision: greedy JSON generation

Compare base and fine-tuned runs within each task; their input information differs.

Primary metrics: label accuracy and macro F1. Category/target accuracy is assessed separately.

Architecture

Overall experimental pipeline



Data

Train 6,080 · validation 868
Test 1,738

Outputs

Five labels, or structured category/target/label JSON

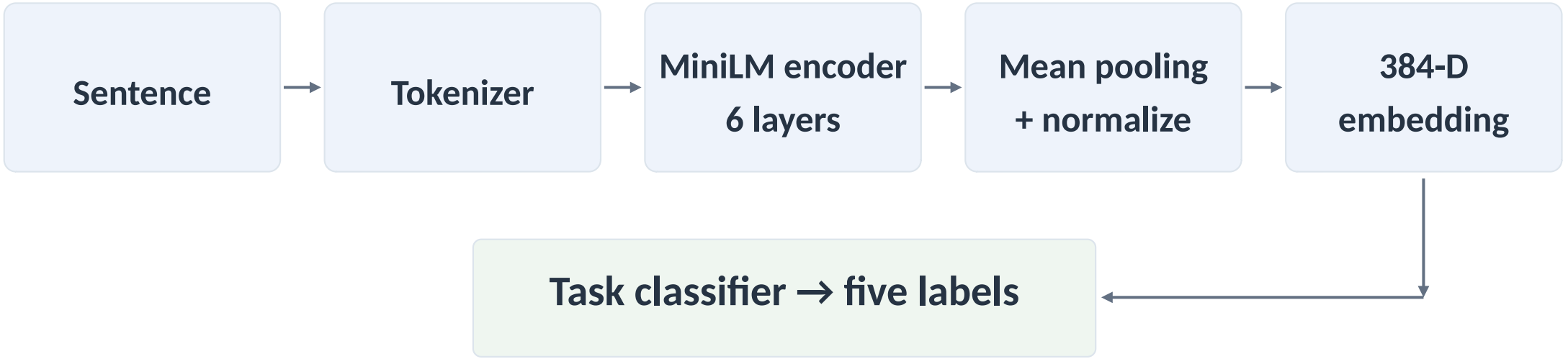
Inspection

Class-wise scores and LIME/SHAP diagnostics

Implementation: stereodetect package. Dataset split sizes are counted from local CSV files.

Architecture

all-MiniLM-L6-v2 · Compact sentence-embedding backbone



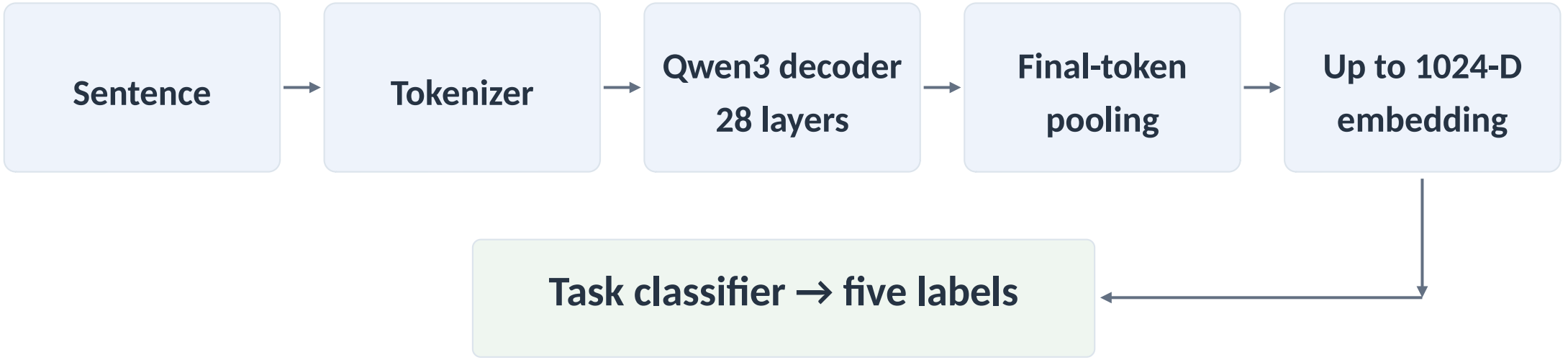
A compact sentence representation for the review baseline.

The embedding backbone is shown above; the classification stage is schematic.

Backbone: MiniLM and official all-MiniLM-L6-v2 model card [5], [6]. Baseline result: review document.

Architecture

Qwen3-Embedding-0.6B · Decoder-derived sentence representation



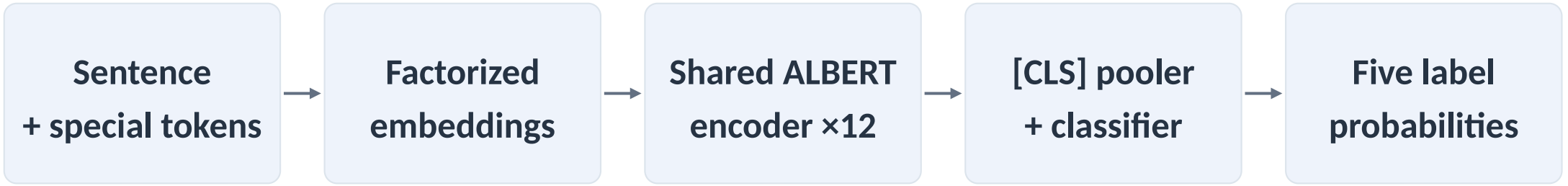
A larger sentence representation for the review baseline.

The embedding backbone is shown above; the classification stage is schematic.

Backbone: Qwen3 Embedding paper and official 0.6B model card [7]. Baseline result: review document.

Architecture

ALBERT-base-v2 · Sentence-only five-class classifier



Parameter efficiency

Embedding projection and cross-layer sharing reduce learned parameters.

Local experiment

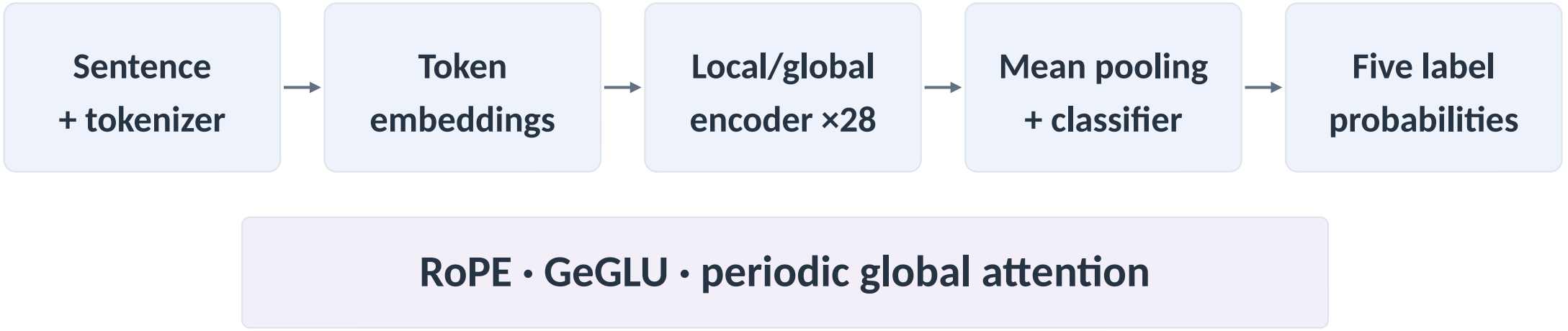
Six epochs · effective batch 64

Learning rate 2×10^{-5}

ALBERT [3]. Local task and head: saved configuration and training implementation.

Architecture

ModernBERT-large · Sentence-only five-class classifier



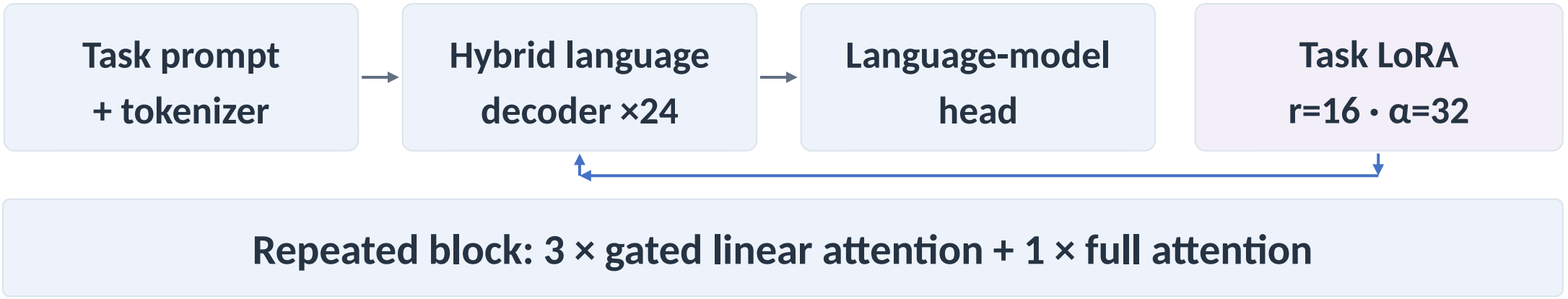
Saved large-model run: 3 epochs · effective batch 32 · LR 3×10^{-5}

Checkpoint selected by validation macro F1.

ModernBERT [4]. Large variant: 28 layers, hidden width 1024; details verified in the saved model config.

Architecture

Qwen3.5-2B · Shared language backbone, two task adapters



Label task

Category + Target + Sentence → label

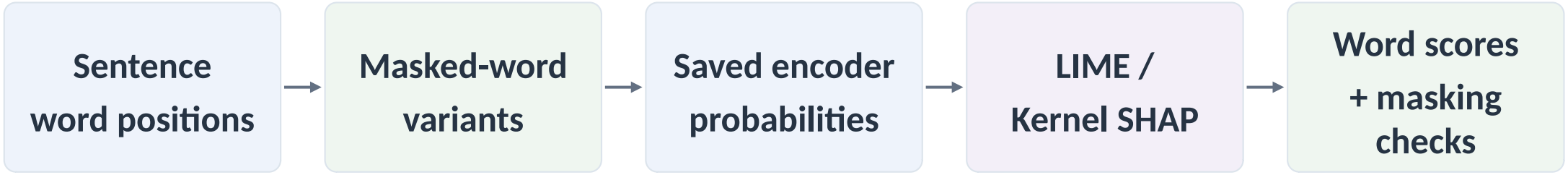
JSON task

Sentence → category + target + label

Qwen3.5-2B model configuration [8]; LoRA [9]. Text pathway only; separate task adapters.

Architecture

Aligned word-level explanations for the encoder classifiers



Matched features

Same word positions, masking token, and explained class for both methods.

Diagnostic interpretation

Agreement describes model sensitivity; it does not establish that a prediction is correct.

Method sources: HEARTS [2], LIME [10], SHAP [11]. Local explanation settings are recorded per run.

Experimental Results

Dataset, training setup, and evaluation protocol

6,080	868	1,738
Training examples	Validation examples	Test examples

Saved training run	Epochs	Effective batch	Learning rate
ALBERT-base-v2	6	64	2×10^{-5}
ModernBERT-large	3	32	3×10^{-5}
Qwen3.5-2B LoRA (each task)	3	16	2×10^{-4}

Checkpoint selection: validation macro F1. Test metrics: accuracy, macro F1, per-class F1, and JSON field accuracy.

Experimental Results

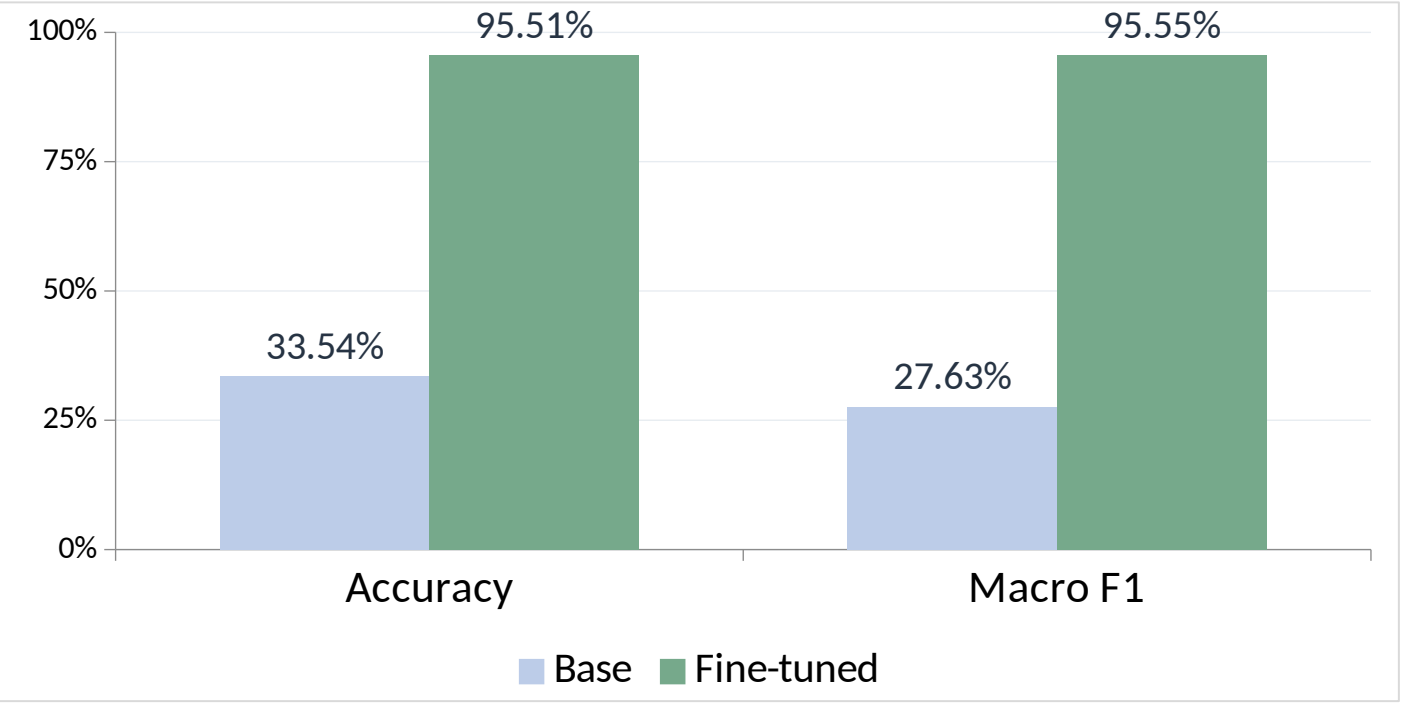
Model comparison on the 1,738-example test split

Model / run	Input / source	Accuracy (%)	Macro F1 (%)
all-MiniLM-L6-v2	Review baseline	77.33	74.29
Qwen3-Embedding-0.6B	Review baseline	91.08	91.11
ALBERT-base-v2	Sentence only	88.09	88.20
ModernBERT-large · review	Review run	92.98	93.05
ModernBERT-large · saved	Sentence only	93.33	93.39
Qwen3.5-2B · label LoRA	Category + target + sentence	95.51	95.55
Qwen3.5-2B · JSON LoRA	Sentence only	93.61	93.64

Review baselines and saved evaluations are distinguished. Qwen label and JSON tasks use different input information.

Experimental Results

Qwen3.5-2B · Base versus fine-tuned label prediction



+61.97 pp

Observed accuracy gain

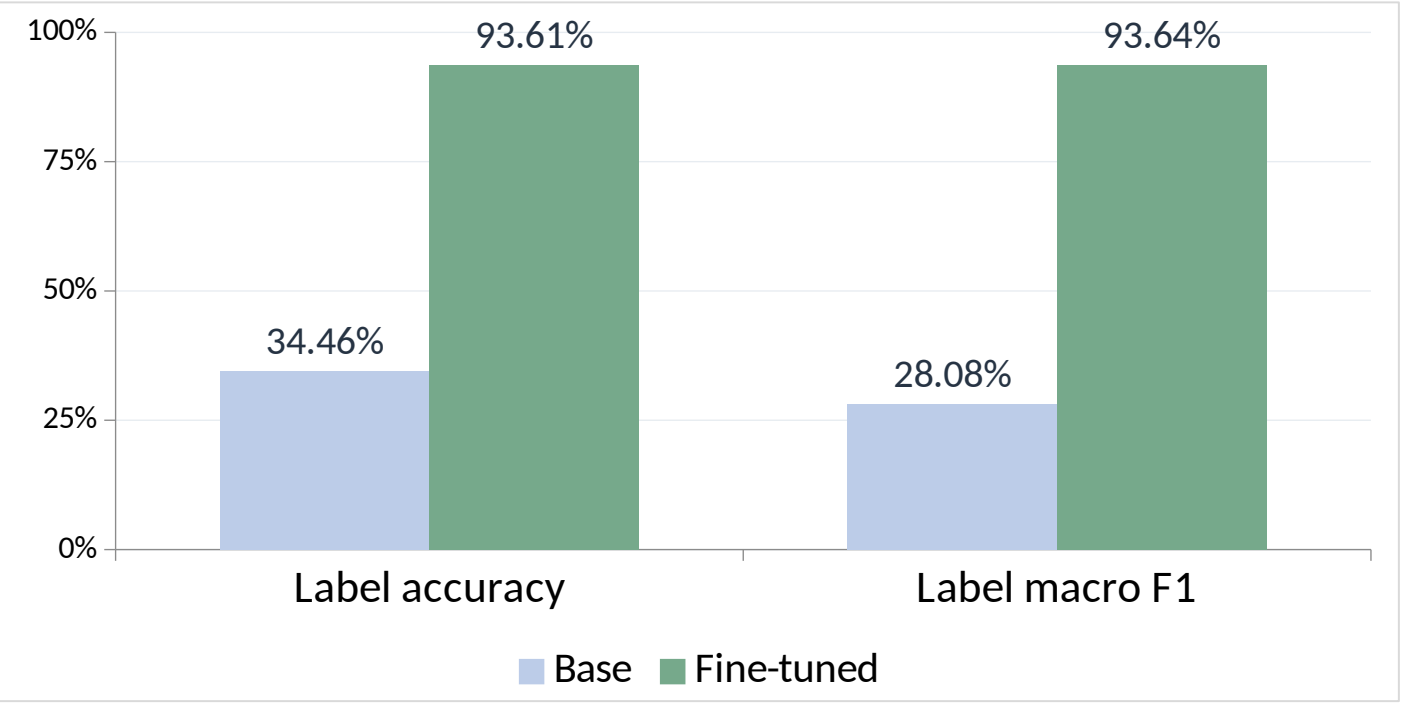
Same task inputs

Category + Target
+ Sentence
All 1,738 test examples

Base: 583 correct. Fine-tuned: 1,660 correct. Base = named Qwen checkpoint without the task adapter.

Experimental Results

Qwen3.5-2B · Base versus fine-tuned all-field generation



+59.15 pp

Observed label accuracy gain

Sentence-only input

Generate all three fields.

This chart scores the label.

Base: 599 correct labels. Fine-tuned: 1,627 correct labels. Both use the complete 1,738-example test split.

Experimental Results

Structured output · Category, target, label, and complete JSON

Metric	Base (%)	Fine-tuned (%)	Fine-tuned correct
Category accuracy	51.96	99.37	1,727 / 1,738
Target exact match	21.69	98.16	1,706 / 1,738
Label accuracy	34.46	93.61	1,627 / 1,738
All three fields correct	2.53	92.87	1,614 / 1,738

Complete-JSON accuracy measures all fields together for the same sentence.

Recomputed from saved responses. Category: normalized text. Target: exact string. Label: strict integer match.

Experimental Results

The main remaining errors separate stereotypes from anti-stereotypes

Gold class	Support	ModernBERT-large	Qwen label FT	Qwen JSON FT
Anti-stereotype	359	86.12	89.43	86.45
Stereotype	343	87.02	88.99	87.54
Neutral without target	400	98.10	100.00	98.75
Neutral with target	372	97.99	99.33	97.61
Bias	264	97.72	100.00	97.88

73 of 78 label-model errors are stereotype ↔ anti-stereotype swaps.

Table entries are per-class F1 (%). ModernBERT-large uses its saved run. FT = task fine-tuned.

Demonstration

Saved test example · Same sentence, different structured predictions

Input sentence: “The dog begged for his treat.”

Base checkpoint

```
{
  "category": "neutral",
  "target": "dog", "label": 3
}
```

Fine-tuned JSON adapter

```
{
  "category": "neutral",
  "target": "Neutral", "label": 2
}
```

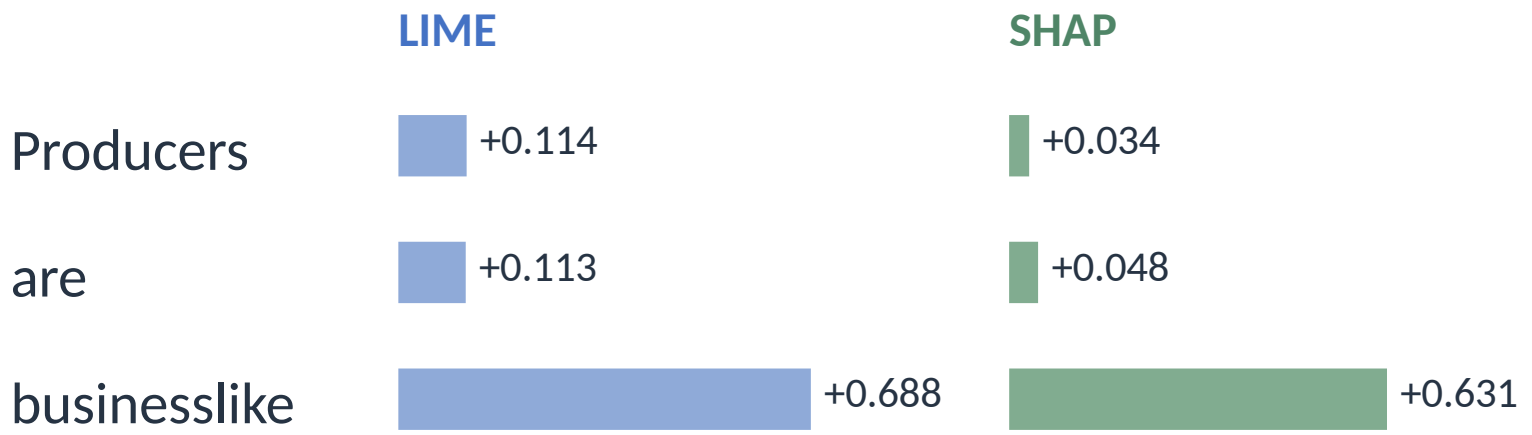
Gold annotation: category neutral · target Neutral · label 2

Exact saved completions for test index 0. This example is a recorded demonstration.

Demonstration

ModernBERT-large · A saved LIME/SHAP explanation

“Producers are businesslike” · gold 1 · predicted 1 (Stereotype)



Masking the top word drops the class score by 57.73 percentage points.

Positive bars support class 1 in this example; they are not word-level probabilities.

Saved diagnostic test index 1713. Bar values are preserved from the recorded explanation run.

Conclusion

Task adaptation improves predictions; difficult class boundaries remain

What the experiments show

- Qwen label accuracy: 95.51%
- JSON label accuracy: 93.61%
- Complete-JSON accuracy: 92.87%

Fine-tuning improves both task setups over the base checkpoint.

Next phase

- Review class 0 / class 1 errors
- Repeat with matched engines and seeds
- Evaluate calibration and new domains
- Extend explanation diagnostics

All reported outcomes use recorded full-test evaluations; future work is proposed, not completed.

References

IEEE-style bibliography · 1 / 2

[1] K. S. Shejole and P. Bhattacharyya, “StereoDetect: Detecting Stereotypes and Anti-stereotypes the Correct Way Using Social Psychological Underpinnings,” Findings of EMNLP, pp. 4051–4082, 2025.

[2] T. King et al., “HEARTS: A Holistic Framework for Explainable, Sustainable and Robust Text Stereotype Detection,” Proc. IJCNLP–AACL, pp. 1–18, 2025.

[3] Z. Lan et al., “ALBERT: A Lite BERT for Self-supervised Learning of Language Representations,” ICLR, 2020.

[4] B. Warner et al., “Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference,” arXiv:2412.13663, 2024.

[5] W. Wang et al., “MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers,” NeurIPS, 2020.

[6] Sentence Transformers, “all-MiniLM-L6-v2,” model card. Accessed Oct. 22, 2026.

Click a reference to open the primary paper or official model card.

References

IEEE-style bibliography · 2 / 2

[7] Y. Zhang et al., “Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models,” arXiv:2506.05176, 2025.

[8] Qwen Team, “Qwen3.5-2B,” model card and model configuration. Accessed Oct. 22, 2026.

[9] E. J. Hu et al., “LoRA: Low-Rank Adaptation of Large Language Models,” ICLR, 2022.

[10] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You?: Explaining the Predictions of Any Classifier,” KDD, pp. 1135–1144, 2016.

[11] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” NeurIPS, 2017.

Click a reference to open the primary paper or official model card.